

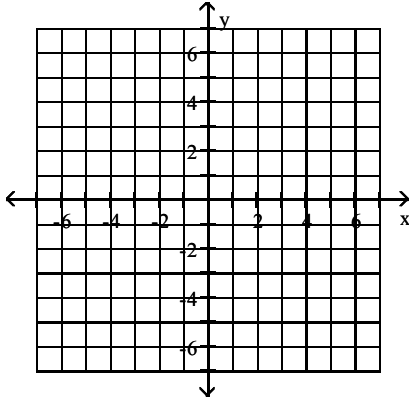
Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Plot the given point in a rectangular coordinate system.

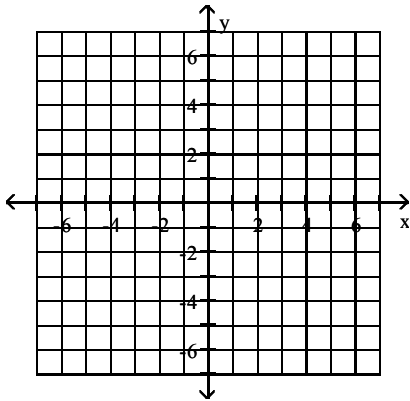
1) $(2, -1)$

1) _____



2) $(-2, -6)$

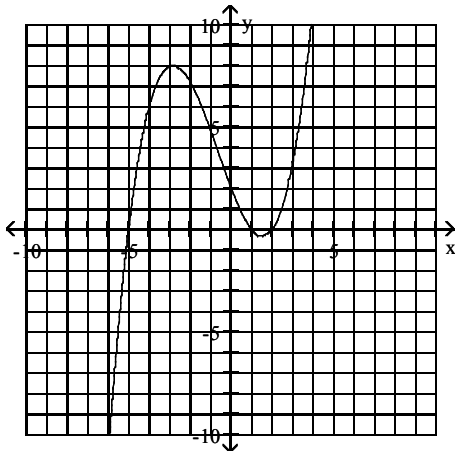
2) _____



Use the graph to determine the x- and y-intercepts.

3)

3) _____



Solve and check the linear equation.

4) $4x - 6 = 30$

4) _____

5) $7x + 6 = -10 + 9x$

5) _____

6) $-6x + 7 - 4(x + 1) = 3x + 1$

6) _____

Solve the equation.

7) $\frac{x}{2} = \frac{x}{8} + 3$

7) _____

8) $\frac{x - 14}{7} + \frac{x + 2}{2} = x - 9$

8) _____

First, write the value(s) that make the denominator(s) zero. Then solve the equation.

9) $\frac{10}{x} = \frac{3}{2x} + 34$

9) _____

Use the five-step strategy for solving word problems to find the number or numbers described in the following exercise.

10) When 3 times a number is subtracted from 7 times the number, the result is 36. What is the number? 10) _____

11) When 70% of a number is added to the number, the result is 204. What is the number? 11) _____

Solve the problem.

12) A car rental agency charges \$200 per week plus \$0.10 per mile to rent a car. How many miles can you travel in one week for \$270? 12) _____

Solve the formula for the specified variable.

13) $d = rt$ for t 13) _____

Solve the equation by factoring.

14) $x^2 = x + 20$

14) _____

15) $6x^2 - 17x = 3$

15) _____

Solve the equation by the square root property.

16) $5x^2 = 80$

16) _____

17) $3(x - 2)^2 = 15$

17) _____

Solve the equation using the quadratic formula.

18) $x^2 + 7x + 7 = 0$

18) _____

19) $7x^2 = -12x - 3$

19) _____

Solve the polynomial equation by factoring and then using the zero product principle.

20) $x^3 + 5x^2 + 6x = 0$

20) _____

Solve the radical equation, and check all proposed solutions.

21) $\sqrt{x + 4} = 6$

21) _____

$$22) \sqrt{4x + 12} = x$$

22) _____

Solve the absolute value equation or indicate that the equation has no solution.

$$23) |x + 1| = 8$$

23) _____

$$24) |7x - 5| + 2 = -7$$

24) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Determine whether the equation is an identity, a conditional equation, or an inconsistent equation.

$$25) 5(4x - 12) = 20x - 60$$

A) Identity

B) Conditional equation

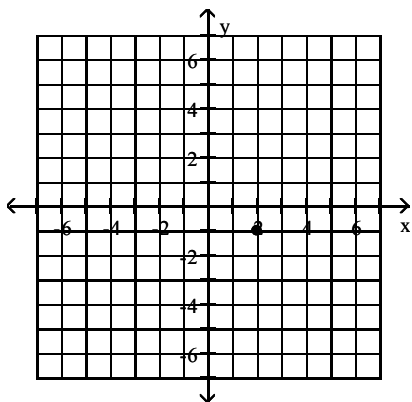
C) Inconsistent equation

25) _____

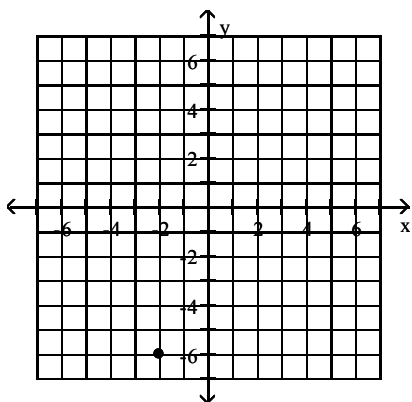
Answer Key

Testname: MTH_131_EXAM2_FALL_2012B

1)



2)



3) x-intercepts: 2, 1, -5; y-intercept: 2

4) {9}

5) {8}

6) $\left\{\frac{2}{13}\right\}$

7) {8}

8) $\left\{\frac{112}{5}\right\}$

9) $x \neq 0; \left\{\frac{1}{4}\right\}$

10) 9

11) 120

12) 700 miles

13) $t = \frac{d}{r}$

14) {-4, 5}

15) $\left\{-\frac{1}{6}, 3\right\}$

16) {-4, 4}

17) $\{2 \pm \sqrt{5}\}$

18) $\left\{\frac{-7 - \sqrt{21}}{2}, \frac{-7 + \sqrt{21}}{2}\right\}$

Answer Key

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19) $\left\{ \frac{-6 - \sqrt{15}}{7}, \frac{-6 + \sqrt{15}}{7} \right\}$

20) $\{0, -2, -3\}$

21) $\{32\}$

22) $\{6\}$

23) $\{-9, 7\}$

24) $\emptyset$

25) A